

SOUVIK NATH

Software Engineer | AI Applications
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PROFESSIONAL SUMMARY

Full Stack Engineer with hands-on experience developing scalable web applications and backend systems using Next.js, React.js, TypeScript, Django, and PostgreSQL. Proven experience building secure REST APIs, authentication systems, and high-performance data-driven applications. Skilled in Docker, CI/CD, and modern software engineering practices. Passionate about AI-powered products, currently building SaaS applications using OpenAI, RAG, semantic search, embeddings, and vector databases to create intelligent user experiences.

TECHNICAL SKILLS

- **Frontend:** Next.js, React.js, TypeScript, JavaScript, Tailwind CSS, HTML5, CSS3
- **Backend:** Python, Django, Django REST Framework, REST APIs, JWT Authentication, RBAC
- **Databases:** PostgreSQL, MySQL, pgvector
- **AI/ML:** Generative AI, LLM Applications, OpenAI API, RAG, Semantic Search, Embeddings, LangChain, Vector Search
- **DevOps & Cloud:** Docker, GitHub Actions, CI/CD, Vercel, AWS
- **Tools:** Git, GitHub, Postman

PROFESSIONAL EXPERIENCE

Gowdanar Technosoft Pvt. Ltd. — Software Engineer
Bangalore, Karnataka, India

12/2023 — Present

- Engineered scalable frontend modules using Next.js and React.js, reducing page load times by ~30% and significantly improving dashboard responsiveness.
- Built and optimized RESTful APIs using Django REST Framework, improving backend workflow efficiency and reducing average API response time by ~25% across compliance modules.
- Implemented JWT authentication and RBAC systems for three user roles, fortifying application security and access management, and processing 75,000+ authentication requests daily with zero downtime.
- Integrated React/Next.js frontend systems with PostgreSQL and MySQL databases, improving data retrieval performance and increasing API response consistency by ~30%.
- Delivered production-ready features in agile sprint cycles, accelerating release timelines by ~20% while reducing recurring defects through proactive code reviews and testing.
- Designed scalable backend workflows for audit trails and compliance reporting, reducing repetitive manual processes by ~40% and improving maintainability.

PROJECTS

1. AI-Powered Personal Finance Assistant (RAG + Semantic Search Platform)

Tech Stack: Next.js, TypeScript, Django REST Framework, PostgreSQL, pgvector, REST APIs, OpenAI, RAG, Embeddings, JWT

- Built and deployed an AI-powered personal finance platform using Next.js, TypeScript, Django REST Framework, PostgreSQL, and pgvector for financial document analysis and transaction management.
- Architected secure REST APIs and JWT authentication for protected access to financial data and seamless frontend-backend integration.
- Developed automated PDF and CSV transaction processing pipelines, extracting, normalizing, and categorizing financial records for spending analysis and financial reporting.
- Leveraged OpenAI embeddings, semantic search, and Retrieval-Augmented Generation (RAG) to deliver AI-driven expense categorization and financial insights.
- Engineered vector search and transaction retrieval workflows using embeddings and similarity search, improving contextual accuracy for AI-generated financial recommendations.

Live Demo: ai-finance-assistant-phi.vercel.app | **GitHub:** github.com/souviknath18/ai-finance-assistant

2. AI Career Coach Platform (LLM-Powered Career Guidance System)

Tech Stack: TypeScript, Django REST Framework, PostgreSQL, OpenAI API, pgvector

- Built an AI-powered career coaching platform for resume analysis, job matching and personalized career guidance.
- Constructed secure REST APIs and JWT authentication to support user management and AI-powered workflows.
- Integrated RAG, OpenAI embeddings, semantic search, and pgvector to deliver context-aware career recommendations.
- Created an AI chat assistant and mock interview system using LLMs to provide personalized career coaching and interview feedback.

Live Demo: ai-career-coach-phi.vercel.app | **GitHub:** github.com/souviknath18/ai-career-coach

EDUCATION

Bachelor of Technology — Electronics and Communication Engineering, 09/2021
Assam University, Silchar

LANGUAGES

- English
- Hindi
- Bengali